

22PH202 ELECTROMAGNETISM AND MODERN PHYSICS

UNIT III – LM 3. 2 - - ELECTROMAGNETIC ENERGY

1.1 Anatomy of an Electromagnetic Wave

Energy, a measure of the ability to do work, comes in many forms and can transform from one type to another. Examples of stored or potential energy include batteries and water behind a dam. Objects in motion are examples of kinetic energy. Charged particles—such as electrons and protons—create electromagnetic fields when they move, and these fields transport the type of energy we call electromagnetic radiation, or light.



Figure-1 Electromagnetic Wave

1.2 Electromagnetic and Mechanical waves

Mechanical waves and electromagnetic waves are two important ways that energy is transported in the world around us. Waves in water and sound waves in air are two examples of mechanical waves. Mechanical waves are caused by a disturbance or vibration in matter, whether solid, gas, liquid, or plasma. Matter that waves are traveling through is called a medium. Water waves are formed by vibrations in a liquid and sound waves are formed by vibrations in a gas (air). These mechanical waves travel through a medium by causing the

molecules to bump into each other, like falling dominoes transferring energy from one to the next. Sound waves cannot travel in the vacuum of space because there is no medium to transmit these mechanical waves as shown in figure 2



Figure-2 Representation of Electromagnetic Wave

1.3 ELECTROMAGNETIC WAVES

Electricity can be static, like the energy that can make your hair stand on end. Magnetism can also be static, as it is in a refrigerator magnet. A changing magnetic field will induce a changing electric field and vice-versa—the two are linked. These changing fields form electromagnetic waves. Electromagnetic waves differ from mechanical waves in that they do not require a medium to propagate. This means that electromagnetic waves can travel not only through air and solid materials, but also through the vacuum of space.

In the 1860's and 1870's, a Scottish scientist named James Clerk Maxwell developed a scientific theory to explain electromagnetic waves. He noticed that electrical fields and magnetic fields can couple together to form electromagnetic waves. He summarized this relationship between electricity and magnetism into what are now referred to as "Maxwell's Equations."

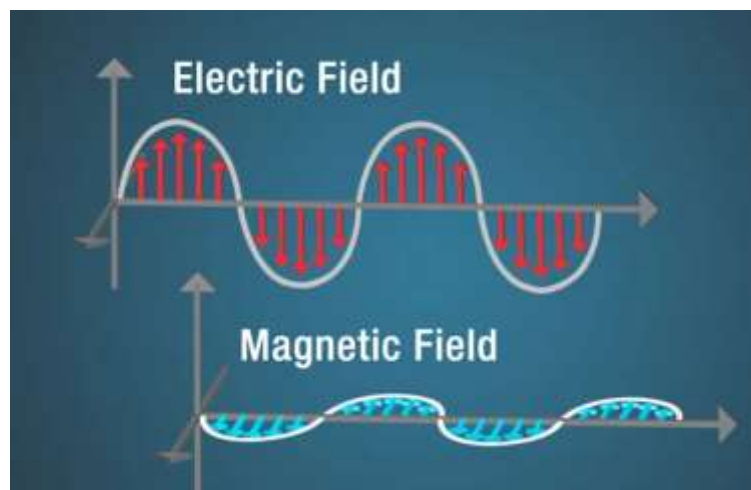


Figure-3 Electromagnetic Wave

1.4 ELECTROMAGNETIC RADIATION

From the Figure 4 ,Electromagnetic radiation can be defined as a form of energy that is generated when electrically charged particles move through matter or a vacuum. Alternatively, it can also be produced by oscillating magnetic and electric disturbances.

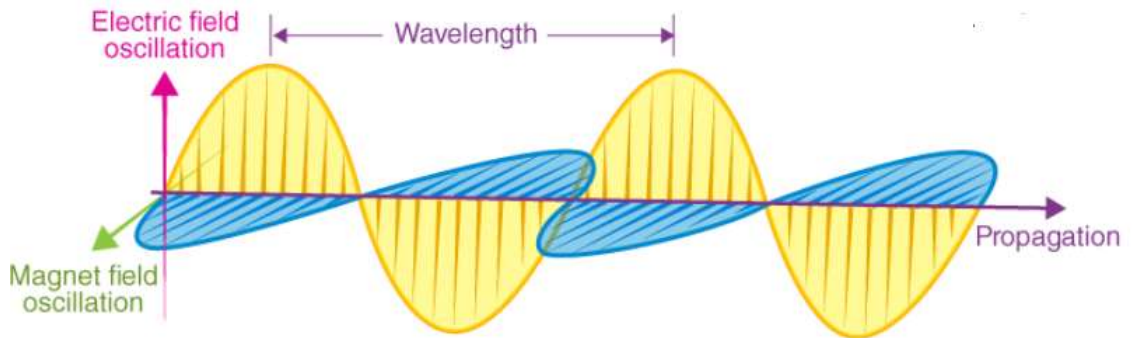


Figure-4 Electromagnetic Radiation

- **Energy from Charged Particles:** Electromagnetic radiation is generated when electrically charged particles, such as electrons, move through a medium or empty space. These particles carry energy as they travel, creating a wave-like pattern of electric and magnetic fields.
- **Electric and Magnetic Fields:**Electromagnetic radiation consists of electric and magnetic fields. These fields are intimately connected and perpendicular to each other, with a 90-degree angle. As the combined waves propagate through space, they move perpendicular to the electric and magnetic oscillations occurring during the disturbance.
- **Diverse Forms of Electromagnetic Radiation:** Electromagnetic radiation encompasses a vast spectrum of energy, ranging from low-energy radio waves to high-energy gamma rays. Some common forms of electromagnetic radiation include microwaves, television waves, radio waves, infrared radiation, visible light, ultraviolet radiation, X-rays, and gamma rays. Each form has different wavelengths and energy levels, resulting in unique properties and applications.
- **Transverse Waves:** Electromagnetic radiation travels as [transverse waves](#), meaning that the oscillations occur perpendicular to the direction of wave propagation. This characteristic allows electromagnetic radiation to traverse space, enabling it to propagate through a vacuum, unlike other types of waves.

1. 4.1 Properties of Electromagnetic Radiation

- **Photons:** Electromagnetic radiation is emitted as discrete packets of energy called photons. These photons carry light energy and travel at the constant speed of light.

- **Quantized Harmonic Waves:** Electromagnetic radiation can be described as quantized harmonic waves. These waves exhibit specific frequencies and energy levels.
- **Categorization by Wavelength:** Electromagnetic radiation is grouped into different categories based on its wavelength within the electromagnetic spectrum. Each category has distinct properties and applications.
- **Perpendicular Electric and Magnetic Fields:** Electromagnetic waves consist of perpendicular electric and magnetic fields. These fields oscillate in directions that are perpendicular to each other and to the direction of wave propagation.
- **Wavelength, Amplitude, and Frequency:** Electromagnetic radiation exhibits properties such as wavelength, amplitude, and frequency. Wavelength (λ) represents the distance between two adjacent points in phase, while frequency determines the number of wave cycles per second. Amplitude refers to the maximum displacement of the wave from its equilibrium position.
- **Travel through Empty Space:** Unlike other types of waves, electromagnetic radiation can travel through empty space. It does not require a medium such as a solid, liquid, or gas for propagation.
- **Constant Speed of Light:** The speed of light in a vacuum, denoted as "c," is a fundamental constant of approximately 2.99792458×10^8 metres per second. This speed remains constant for all forms of electromagnetic radiation.

THE ELECTROMAGNETIC SPECTRUM

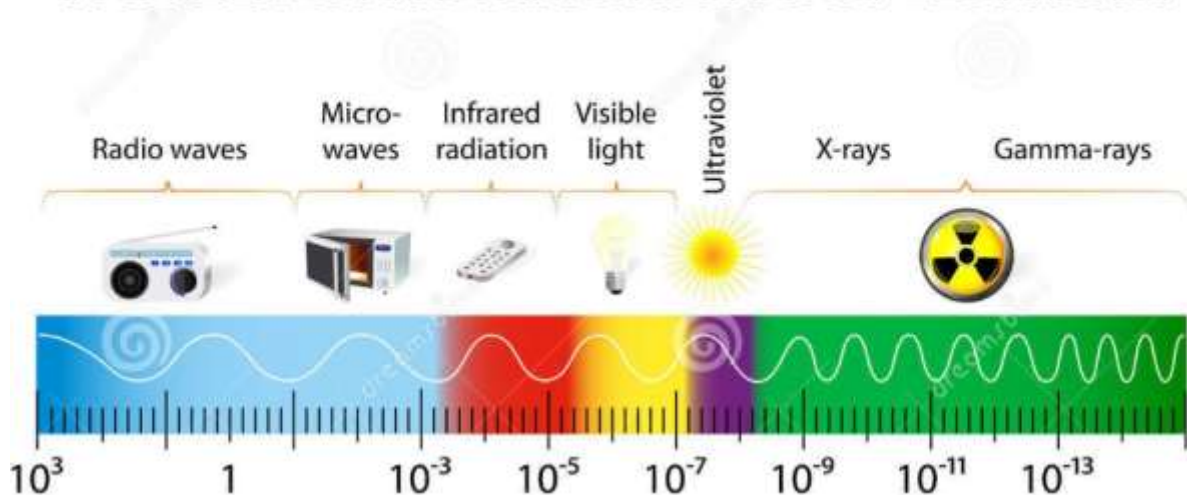


Figure-5 Electromagnetic Spectrum

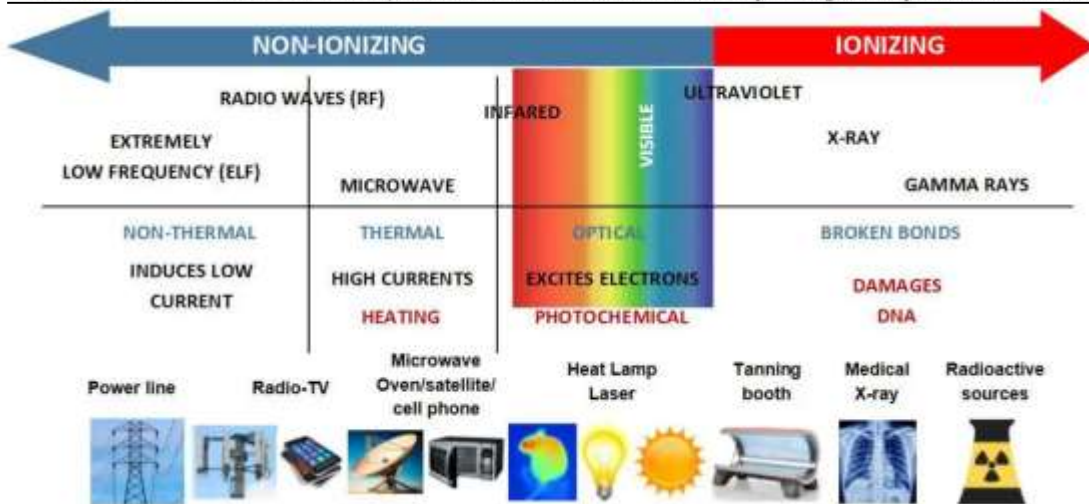


Figure-5 Identification Electromagnetic Spectrum

Type	Wavelength range	Production	Detection
Radio	> 0.1 m	Rapid acceleration and decelerations of electrons in aerials	Receiver's aerials
Microwave	0.1m to 1 mm	Klystron valve or magnetron valve	Point contact diodes
Infra-red	1mm to 700 nm	Vibration of atoms and molecules	Thermopiles Bolometer, Infrared photographic film
Light	700 nm to 400 nm	Electrons in atoms emit light when they move from one energy level to a lower energy level	The eye Photocells Photographic film
Ultraviolet	400 nm to 1nm	Inner shell electrons in atoms moving from one energy level to a lower level	Photocells Photographic film
X-rays	1nm to 10^{-3} nm	X-ray tubes or inner shell electrons	Photographic film Geiger tubes Ionisation chamber
Gamma rays	$<10^{-3}$ nm	Radioactive decay of the nucleus	-do-

1.5 visible light spectrum

The visible light spectrum is the segment of the electromagnetic spectrum that the human eye can view. More simply, this range of wavelengths is called visible light. Typically, the human eye can detect wavelengths from 380 to 700 nanometers.

1. 5.1 Wavelengths Of Visible Light

All electromagnetic radiation is light, but we can only see a small portion of this radiation—the portion we call visible light. Cone-shaped cells in our eyes act as receivers tuned to the wavelengths in this narrow band of the spectrum. Other portions of the spectrum have wavelengths too large or too small and energetic for the biological limitations of our perception. As the full spectrum of visible light travels through a prism, the wavelengths separate into the colors of the rainbow because each color is a different wavelength. Violet has the shortest wavelength, at around 380 nanometers, and red has the longest wavelength, at around 700 nanometers.